



Controllable generative image steganography based on denoising diffusion implicit model

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Abstract

Generative steganography has gained significant attention due to its cover-independent nature, which allows it to evade a broad range of steganalysis techniques. However, existing methods often suffer from poor extraction accuracy at high embedding rates and limited robustness to real-world noise. To address these limitations, we propose the Controllable Generative Image Steganography (CGIS) framework, introducing two bidirectional mapping mechanisms: the Resampling-based RS mapping and the Normal Distribution-based ND mapping. These significantly enhance extraction accuracy even at high embedding rates. Unlike prior stochastic or training-dependent approaches, CGIS utilizes a training-free and deterministic diffusion process, fundamentally distinguishing it from existing methods and ensuring superior robustness. Additionally, a Gaussian Mixture Model-based image denoiser (GMM-ID) ensures the model's resilience to network noise and compression-induced distortion. Experimental results demonstrate that CGIS outperforms existing methods in terms of extraction accuracy (97.75%), detection resistance (96.23%), and image distortion (FID: 2.36), demonstrating superior performance for practical deployment.

Keywords Image steganography · Diffusion model · Generative steganography · Bidirectional mapping schemes · Image denoise

1 Introduction

Image steganography is a technique that embeds secret information into a cover image to make the information undetectable by third parties. It aims to ensure the security and covertness of communication (Tian et al. 2024; Liu et al. 2022). Traditional image steganography methods involve modifying pixels or coefficients in the spatial or frequency domain of the cover image, such as changing the least significant bit (LSB) value or using the syndrome trellis coding (STC) framework (Filler et al. 2011; Li et al. 2014; Rajan and Ganesh 2025). Recently, neural network-based methods have emerged, utilizing neural networks' learning and adaptation abilities to map cover images to stego images,

thereby hiding secret information more covertly (Zhou et al. 2022). For instance, Tang et al. (2017) utilized pixel modification probability maps to guide data embedding. Chu et al. (2017) applied CycleGAN (Zhu et al. 2017) to hide the source image in the generated image imperceptibly. However, these methods require significant prior knowledge and may induce statistical or visual anomalies, thereby increasing steganalysis detectability (Qian et al. 2015). Detection of these anomalies renders the communication ineffective (Li et al. 2023; Ma et al. 2023). In response to these limitations, generative models have been proposed to hide information within the generated image rather than overlaying it.

Generative steganography methods can be categorized into two types: GAN-based and Flow model-based (Li et al. 2025). GAN-based methods utilize the image generation capabilities of GANs to encode the secret information as an input or latent variable and generate the stego image using a generator, followed by decoding the information using an extractor, such as GSN (Wei et al. 2022) based on StyleGAN2 (Karras et al. 2020). Flow model-based methods exploit the invertibility and density estimation capabilities of Flow models to encode the information as a latent variable and generate the stego image through forward transform, followed by

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decoding the information through inverse transform, e.g., RIIS (Xu et al. 2022). However, these methods still exhibit limitations, such as low image quality and limited hidden capacity attributed to the latent variable dimension, potentially leading to data loss or extraction errors, failing to meet the transmission accuracy demand for extensive data in the information age (Zhou et al. 2023).

As a result, diffusion models (Kang et al. 2025) have garnered attention from scholars in the field of steganography. These models, capable of generating and recovering images from random noise without the need for pre-selected vectors or training, offer a novel approach to steganography (Dhariwal and Nichol 2021; Ho et al. 2020). These models are recognized for being more robust and accurate than existing generative models, especially in handling large information capacities. Yu et al. (2024) proposed the CRoSS framework, utilizing the diffusion model to facilitate the transformation between two images for coverless steganography. However, this method exhibits limitations regarding ciphertext, semantic relevance, and application scenarios. Peng et al. (2023) developed the StegaDDPM. This method conceals the secret message in the generated image through sampling from the probability distribution of the diffusion model's inverse process. However, this method lacks robustness to network distortion and JPEG compression, relying on the mean and variance of the stego noise. Recently, Xiaoxiao et al. (2024) have proposed a secure steganographic approach that integrates stable diffusion and a mapping module designed with double-key technology. This approach enables the successful conversion of any secret information into Gaussian information and supports inverse mapping, thereby effectively enhancing the extraction accuracy under lossy conditions.

The overarching challenge is to design a steganographic method that (1) handles high payloads without sacrificing image quality, (2) remains robust under real-world transmission distortions (e.g., network noise, compression), and (3) avoids modifications to existing covers—thus preserving cover integrity and resisting traditional steganalysis. Traditional and some contemporary generative steganography approaches either yield insufficient payload capacity or require sophisticated alterations to a cover image, which can be statistically detectable.

To address the aforementioned limitations regarding embedding capacity and robustness, this paper proposes a novel framework termed Controllable Generative Image Steganography (CGIS). Built upon the Denoising Diffusion Implicit Model (DDIM), CGIS innovatively introduces two bidirectional mapping mechanisms—Resampling-based (RS) mapping and Normal Distribution-based (ND) mapping—to transform binary secret messages into latent variables that statistically align with the standard normal distribution. This alignment ensures high-fidelity image generation and precise message extraction even at elevated embedding rates.

Critically, CGIS achieves this high robustness through a training-free and deterministic diffusion process, which fundamentally differs from the stochastic or training-dependent nature of approaches like StegaDDPM (Peng et al. 2023) and Xiaoxiao et al. (2024). Furthermore, to counteract distortions arising from channel noise and compression, we design a Gaussian Mixture Model-based Image Denoiser (GMM-ID) at the receiving end. By modeling the noise degradation as a Markov process, GMM-ID effectively restores the stego images prior to extraction. Consequently, the proposed CGIS framework unifies these components to offer a secure, high-capacity, and robust solution that significantly outperforms existing generative steganography methods.

The key contributions of our CGIS method are as follows:

- We propose a **training-free, cover-independent** generative steganography method based on *Denoising Diffusion Implicit Models (DDIM)*, sustaining high *extraction accuracy* under large embedding rates.
- We introduce **two bidirectional noise $\leftrightarrow$ message mappings**—a *Resampling-based (RS)* mapping and a *Normal-Distribution (ND)* mapping—that preserve latent normality and enable *reversible, controllable* embedding via sub-interval separation and robust mapping.
- We develop a **Gaussian-mixture-based** image denoiser within a **Markov Random Field (MRF)** framework (*GMM-ID*) to enhance robustness against transmission noise and compression artifacts.
- Extensive experiments across multiple datasets and distortion regimes show that our method surpasses state-of-the-art generative steganography in *image quality, embedding capacity, security* (low detectability), and *robustness*.

2 Related work

2.1 Conventional image steganography

Most conventional image steganography techniques are based on modifying existing covers (Baluja 2017). In these methods, stego images are synthesized through manually crafted procedures that encode secret data into specific textures or patterns. Traditional approaches, such as Least Significant Bit (LSB) (Wengrowski and Dana 2019), Pixel Value Difference (PVD) (Sahu et al. 2022), and Histogram Shift (Tsai et al. 2009), are known for their simplicity but suffer from low security and limited resilience to steganalysis. Subsequently, adaptive methods (Li et al. 2014) were introduced to enhance the visual quality of steganography by minimizing embedding-induced distortion, though their data capacity remains somewhat constrained. Similarly,

transform-based approaches (Kadhim et al. 2019), including those utilizing the Discrete Cosine Transform (DCT) (Pevný et al. 2010), do not offer a high data payload. This has led researchers to explore more secure and less detectable methods for embedding information. Recently, deep learning-based solutions have shown great promise in enhancing image steganography. Notably, GANs (Tang et al. 2017) is being used to create container images that minimize distortion (Tang et al. 2019) through various cost functions. Additionally, invertible neural networks (INNs) (Xiao et al. 2020; Lu et al. 2021) have displayed considerable potential in addressing low-level inverse problems, such as denoising, rescaling, and colorization, offering advantages over traditional methods including autoencoders (Bui et al. 2023), GANs (Abdal et al. 2019), and other learning-based architectures (Yang et al. 2020). However, the stego images generated by these methods often appear unnatural and can easily arouse suspicion, posing risks for real-world transmission (Qiu et al. 2023).

2.2 Coverless steganography

Coverless steganography represents an emerging field within image steganography, embedding secret information without modifying the cover. In contrast to traditional image steganography methods that minimize changes to the cover image, coverless steganography ensures no trace of the secret information remains in the cover image, rendering it resistant to conventional steganalysis methods. Initially, researchers shifted their focus to the GAN model, moving away from the traditional use of GANs for generating vector images to employing GANs for the creation of stego images (Wei et al. 2022). This leveraged the adversarial learning properties of GANs to enhance steganography security. Subsequently, scholars extended this innovative approach to the autoencoder model (Liu et al. 2022), successfully implementing the concept. The use of these models marks a significant departure from methods relying on direct image alterations, introducing a new paradigm in secure communication. Yu et al. (2024) recently introduced CRoSS, a framework for coverless steganography based on stable diffusion, utilizing the ability to transform between two images without training to bolster the security and robustness of image steganography. Nevertheless, this approach necessitates a semantic association between the secret key, the public key, the secret image, and the cover image and is vulnerable to the risk of secret key guessing. Furthermore, this approach faces limitations due to specific application scenarios that depend on the semantic content of the original secret image, thereby diminishing its flexibility. Peng et al. (2023) put forward StegaDDPM, a DDPM-based framework, that capitalizes on the probability distribution between the intermediate state

of the diffusion model's inverse process and the generated image, to map the secret information onto the generated image. However, StegaDDPM demonstrates a lack of robustness in practical scenarios, including network transmission distortion and JPEG compression. Subsequently, Xiaoxiao et al. (2024) present a novel robust generative image steganography scheme via Stable Diffusion. It addresses the issues in existing steganography methods and overcomes the challenges of diffusion models. Moreover, a mapping module is proposed to achieve better performance under lossy conditions. Through experiments, it demonstrates superior extraction accuracy, robustness, security, and image quality. However, this scheme also has certain limitations. For instance, in order to enhance robustness, some hiding capacity has been sacrificed. Additionally, the performance might be influenced by text prompts. As can be seen in the extraction accuracy test on the LAION-10K dataset, the text prompt "noise" could potentially introduce biases. Consequently, a more robust and stable architecture is imperative for coverless image steganography.

2.3 Noise mapping methods

The noise mapping method refers to directly mapping the secret data to intermediate states of the diffusion process and utilizing these states to guide image generation. StegaDDPM (Peng et al. 2023) is a diffusion model-based steganography method that shares some conceptual similarities with our CGIS. StegaDDPM defines 4-bit groupings: 00, 01, 10, and 11, and sequentially assigns a reference value to each grouping, which are 0, 0.25, 0.5, and 0.75 respectively. On this basis, a random perturbation within the range of 0 to 0.25 is added, and the mapping is completed according to this rule. However, this method has obvious limitations. It shows a significant lack of robustness when dealing with boundary values. At the same time, this approach also demonstrates relatively low robustness when dealing with the lossy transmission of images. StegaDDPM is typically limited to payloads of around 3.0 bpp.

In contrast, CGIS employs two bidirectional mapping schemes, namely the resample-based noise mapping (RS) and the normal-distribution-based noise mapping (ND). These mapping methods directly transform the secret message into Gaussian noise. This Gaussian noise, carrying the secret information, is then fed into a conditional diffusion model. This approach enables CGIS to achieve payloads of up to 6.0 bpp, significantly higher than StegaDDPM. Additionally, CGIS offers greater adaptability as it can support arbitrary text prompts for image generation. For example, in various experimental scenarios, CGIS's mapping mechanisms preserve data integrity more effectively under different noise levels and embedding rates. When faced with changes

in the embedding environment, CGIS's bidirectional mapping can better ensure that the secret information is accurately embedded and extracted, while StegaDDPM may experience a decline in performance due to its limited flexibility.

2.4 Denoising techniques

Denoising techniques play a fundamental role in enhancing the robustness of image steganography, particularly in real-world applications where transmission noise, compression artifacts, and other distortions can compromise the integrity of stego images. The ability to recover high-fidelity images after degradation is essential for ensuring the accurate extraction of hidden information.

One notable example is Hu et al.'s (2024) Stable Diffusion-based steganography method, which incorporates a denoising step. However, despite its ability to recover images, this method has notable drawbacks, including a low payload capacity (typically below 2.0 bpp) and a high sensitivity to text prompt biases. These biases can interfere with data recovery, as seen in the LAION-10K dataset extraction tests, where certain prompt words, such as "noise," led to extraction errors.

To overcome these limitations, CGIS integrates the Gaussian Mixture Model-based Image Denoiser (GMM-ID), which provides a more accurate modeling of noise distributions. Unlike previous methods, GMM-ID effectively restores image quality even after distortions such as JPEG compression or Gaussian noise addition. This enhancement allows for more robust and reliable hidden message retrieval, even under high-noise conditions. Experimental results demonstrate that CGIS significantly outperforms Hu et al.'s (2024) method, particularly under challenging noise scenarios. For instance, at $\sigma^2 = 20$, CGIS maintains a bit error rate (BER) of 5.46%, whereas Hu et al.'s (2024) method exhibits a higher BER under similar conditions.

2.5 Steganographic workflow

This section details the operational workflow of the proposed framework, illustrating the complete lifecycle of the secret information.

2.5.1 Step-by-Step generation process

Message transformation First, the secret message, which could be in various forms such as a text file, is converted into a binary string. This binary data is then mapped to Gaussian noise. We use one of the two bidirectional mapping schemes, the resample-based noise mapping (RS) or the normal-distribution-based noise mapping (ND). These mapping methods transform the binary bits into a noise ten-

sor that is compatible with the diffusion model. For example, in the RS method, for a binary ciphertext bit $c = 0$, we sample noise values from a standard normal distribution $N(0, 1)$ until $n > \alpha$ (where α is a predefined threshold), and for $c = 1$, we sample until $n < -\alpha$. This process ensures that the binary data is effectively encoded into a noise pattern that can be used in the subsequent steps.

Image generation The generated noise, combined with a user-defined semantic prompt, is fed into the DDIM. The semantic prompt, such as "a landscape painting" or "a portrait of a person", guides the content of the image to be generated. Over a series of iterations, typically around 300 steps, DDIM refines the noise into a realistic-looking image. During this process, the noise embeds the secret message, while the prompt determines the overall appearance of the image. DDIM operates by iteratively denoising the input noise based on learned patterns. It is an improvement over traditional diffusion models, allowing for faster sampling and more efficient image generation.

Output The final result is a stego image (I') that appears natural to an observer but contains the hidden secret message. This stego image is now ready for transmission.

2.5.2 Extraction process

Denoising After the stego image is transmitted, it may be distorted due to various factors such as noise during network transmission or compression. To address this, we apply the GMM-ID. GMM-ID models the noise and distortion processes as a Markov random process. It uses kernel functions to assess the local similarity of images and preserve edge and texture features. Through an iterative convergence algorithm, it restores the distorted image to a state close to its original form before the transmission-related distortions occurred.

To rigorously assess the computational impact of this module, we analyze its time complexity. Let N denote the total number of pixels in the image (where $N = H \times W$), and K represent the number of iterations performed by the optimization algorithm. In each iteration, the GMM-ID traverses all N pixels to update their pixel values based on the posterior probability calculated from the local neighborhood. Assuming the window size and the number of Gaussian components are constants, the operation per pixel is of constant complexity $\mathcal{O}(1)$. Consequently, the total time complexity of the GMM-ID module is approximately linear with respect to the number of iterations and pixels, denoted as $\mathcal{O}(K \cdot N)$. Although this iterative process introduces an additional time overhead compared to single-pass filtering, it provides the necessary adaptability to complex noise distributions. Given that image steganography extraction is generally not a strict real-time constraint task, this linear increase in computational

cost is an acceptable trade-off for the significant improvement in robustness and the reduction in Bit Error Rate (BER) under varying channel conditions.

Reverse diffusion Following the denoising step by GMM-ID, we employ the deterministic DDIM Inversion process to accurately retrieve the embedded noise. This procedure comprises two distinct stages. First, *Latent Encoding*: the restored container image I_{rec} is mapped from the pixel space back into the latent space using the pre-trained VAE encoder $\mathcal{E}$, yielding the initial latent state $z_0 = \mathcal{E}(I_{rec})$. Second, *Deterministic Inversion*: conditioned on the shared semantic prompt P , we iteratively reverse the diffusion process from timestep $t = 0$ to $t = T$. Unlike the stochastic forward process in standard DDPMs, we leverage the ODE-based approximation of DDIM to deterministically compute the noisy latent z_{t+1} from the current state z_t . Here, the variable z_t denotes the latent state at any intermediate timestep t ($t \in \{0, 1, \dots, T-1\}$), representing the intermediate noisy features as the data is transformed back into noise. The update rule is formulated as follows:

$$z_{t+1} = \underbrace{\sqrt{\bar{\alpha}_{t+1}} \left(\frac{z_t - \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon_\theta(z_t, t, P)}{\sqrt{\bar{\alpha}_t}} \right)}_{\text{predicted } z_0} + \sqrt{1 - \bar{\alpha}_{t+1}} \cdot \epsilon_\theta(z_t, t, P) \quad (1)$$

where $\bar{\alpha}_t$ represents the noise schedule parameter and ϵ_θ denotes the predicted noise by the U-Net. The final latent tensor obtained at timestep T , denoted as z_T , corresponds to the recovered noise tensor N containing the secret message.

Message recovery Finally, the bidirectional mapping that was used to map the binary message to the noise is inverted. If the RS mapping was used during the embedding process, we use the inverse RS mapping (RS^{-1}), and if the ND mapping was used, we use (ND^{-1}). This inverse mapping converts the recovered noise back into the original binary message, completing the extraction process.

2.5.3 Equation contextualization

In this section, we have equations like $I' = g(N, P)$ and related ones for extraction. We now provide more context for these equations. For the equation $I' = g(N, P)$, g represents the generative function implemented by the DDIM. N is the noise tensor that has been mapped from the secret message using the RS or ND method, and P is the semantic prompt that guides the image-generation process. This equation essentially describes how the DDIM takes the noise containing the secret information and the semantic prompt as inputs and generates the stego image.

For the extraction-related equation (for example, if we represent the extraction process as $N = h(I')$), h represents the reverse process. This process retrieves the noise N from the stego image I' for the purpose of message decoding. By clearly defining these equations in the context of the overall steganographic process, we aim to make the mathematical representation more understandable and tie it closely to the practical steps involved.

3 Method

3.1 Overall framework

We introduce an innovative CGIS framework based on the denoising diffusion implicit model for hiding and extracting secret information in images. This framework involves two parties, the sender and the receiver, and relies on three essential techniques: bidirectional mapping of binary messages, GMM-ID, and conditional diffusion model. These components work in a sequential manner in the steganographic process.

First, the secret message is mapped to Gaussian noise using the bidirectional mapping methods detailed in Section 4.2. This mapping process transforms the binary-formatted secret message into Gaussian-like noise, which serves as the cover of the secret information. Then, the noise generated in this step is used as an input to generate a stego image via the conditional diffusion model described in Section 4.4. The conditional diffusion model, based on DDIM, takes the noise and semantic prompts related to the image's application scenario to generate a new container image, effectively embedding the secret message into the generated image.

After the stego image is transmitted, it may be affected by various real-world distortions such as noise and compression. At this point, the GMM-ID denoiser, as presented in Section 4.3, comes into play. The denoiser restores the distorted image to its original or near-original state, which is crucial for enabling accurate message extraction at the receiving end. This can be observed in Fig. 1.

Furthermore, to ensure the deterministic reversibility of the steganographic process, precise synchronization between the sender and receiver is required. We explicitly define the "shared parameters" essential for this framework, which act similarly to a symmetric key in cryptography:

1. **Model Weights:** Both parties must utilize the exact same pre-trained weights of the diffusion model, specifically the U-Net and VAE components of Stable Diffusion, to ensure that the forward generation and reverse inversion processes are mathematically aligned.

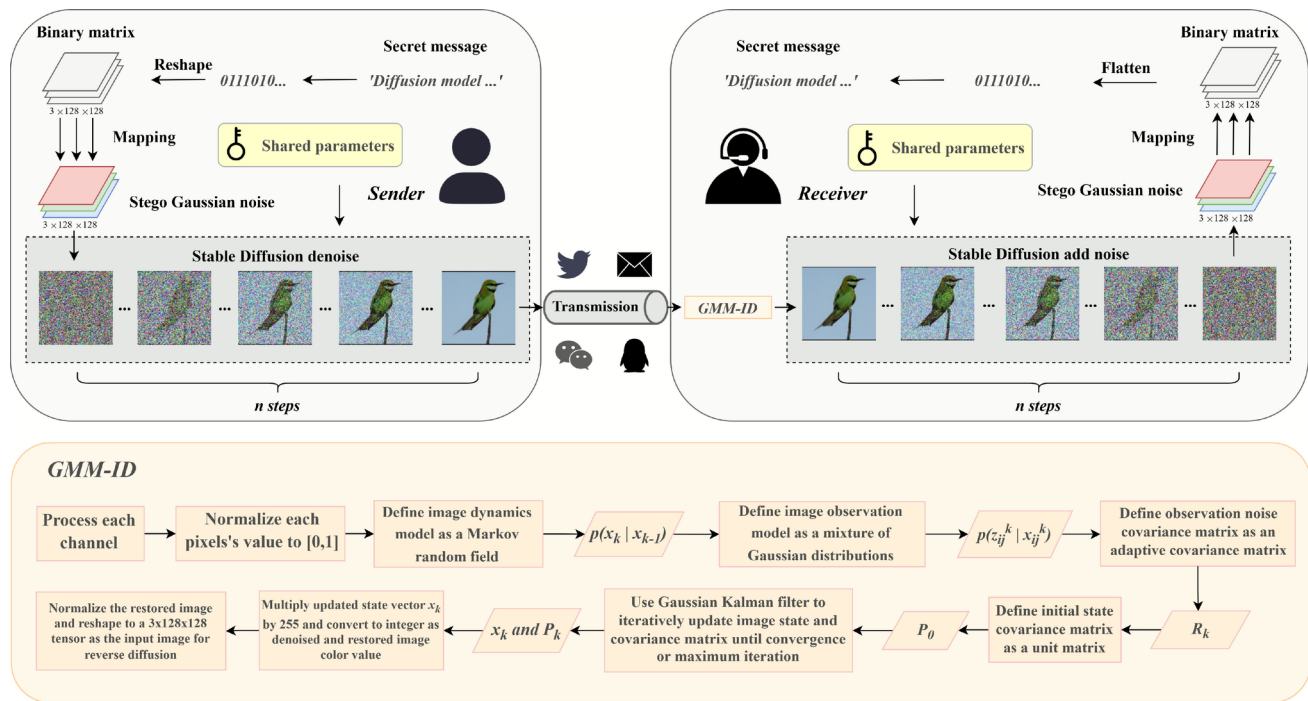


Fig. 1 The visualization of our CGIS framework

- Semantic Prompt P :** As the condition guiding the diffusion process, the text prompt determines the semantic context of the generated image. It must be shared to allow the receiver to reconstruct the identical noise prediction path during the reverse process.
- Mapping Hyperparameters:** These serve as the decoding keys for the bidirectional mapping module. For the RS scheme, the shared parameter is the threshold α ; for the ND scheme, they include the interval parameters such as the range $[a, b]$, count n , and variance σ^2 .

3.2 Bidirectional mapping schemes

To regulate the distribution of encrypted noise post-mapping, we introduce two mapping methods grounded in the concept of interval separation for the diffusion model: the RS and ND methods. The ND method represents an advanced version of the RS method, utilizing a more sophisticated mapping mechanism to ensure a more secure mapping process. Our experimental results indicate that these methods are effective in mapping binary ciphertexts into Gaussian noise. Furthermore, the receiver can employ the inverse noise-to-binary method to retrieve the original ciphertext information from the Gaussian noise, despite the presence of added noise. We have delineated both methods in detail in the following sections.

3.2.1 Noise mapping based on the resample (RS)

Algorithm 1 Binary to noise of RS.

Input: Cipher binary $c \in \{0, 1\}$, threshold α

Output: Noise $n \sim N(0, 1)$

```

1: if  $c = 0$  then
2:   repeat
3:      $n \leftarrow$  Sample a value from  $N(0, 1)$ 
4:   until  $n > \alpha$ 
5: else if  $c = 1$  then
6:   repeat
7:      $n \leftarrow$  Sample a value from  $N(0, 1)$ 
8:   until  $n < -\alpha$ 
9: end if
10: return  $n$ 

```

Algorithm 2 Noise to binary of RS.

Input: Noise $n \sim N(0, 1)$

Output: Binary $c \in \{0, 1\}$

```

1: if  $n > 0$  then
2:    $c \leftarrow 0$ 
3: else if  $n < 0$  then
4:    $c \leftarrow 1$ 
5: end if
6: return  $c$ 

```

We employ a novel bi-directional mapping technique that converts ciphertext c into noise n , closely resembling a Gaussian distribution $N(0, 1)$. To accomplish this, we sample noise values from the distribution until they meet specific criteria derived from the ciphertext values. Specifically, for $c = 0$, we continue sampling n until $n > \alpha$, and for $c = 1$, sampling persists until $n < -\alpha$, where α represents a predefined threshold. In our experimental validation, α was empirically set to 0.01. This setting was chosen based on a trade-off analysis where a larger α increases the separation margin between bits to reduce the BER but concurrently slows down the sampling convergence and distorts the Gaussianity of the latent noise. The value 0.01 provides an optimal balance by minimizing the bit error rate while maintaining fast convergence for the sampling algorithm. This technique establishes separation intervals between noise values corresponding to differing ciphertext values, simultaneously preserving the noise's Gaussian-like characteristics. For decoding the noise values, we apply a straightforward rule: if $n > 0$, it is decoded as $c = 0$; if $n < 0$, as $c = 1$. The concrete operations are shown in the Algorithms 1 and 2.

It should be noted that these mappings are not designed to secure the message against interception; rather, they facilitate the reversible transformation of binary data into a noise format suitable for the diffusion model. The security of CGIS lies not in the mappings but in the generative process itself, where the stego image's natural appearance and lack of a modified cover thwart steganalysis detection. Unlike encryption (e.g., Advanced Encryption Standard), which uses keys to scramble data, RS is a mapping scheme that converts binary bits into scalar noise values using a randomized but reversible process. It is optimized for efficiency and reversibility, not confidentiality.

3.2.2 Noise mapping based on the normal distribution (ND)

Algorithm 3 Binary to noise of ND.

Input: Plaintext m , value range of ciphertext $[a, b]$, number of intervals n , number of bits per interval k , variance of normal distribution σ^2

Output: Ciphertext c

```

1: Compute the length of each interval  $\Delta = \frac{b-a}{n}$ 
2: Convert  $m$  to binary representation  $b_m$ 
3: Divide  $b_m$  into  $k$ -bit segments  $s_1, s_2, \dots, s_t$ 
4: for  $i = 1$  to  $t$  do
5:   Convert  $s_i$  to decimal representation  $d_i$ 
6:   Compute the index of the interval that contains  $d_i$  as  $j_i = \lfloor \frac{d_i - a}{\Delta} \rfloor + 1$ 
7:   Compute the centre of the  $j_i$ th interval as  $\mu_{j_i} = a + (j_i - 0.5)\Delta$ 
8:   Generate a random variable  $y_i \sim N(\mu_{j_i}, \sigma^2)$  as the mapping value of  $d_i$ 
9: end for
10: Concatenate  $y_1, y_2, \dots, y_t$  as the ciphertext  $c$ 
11: return  $c$ 

```

On the basis of the RS method, we introduce a second mapping method that preserves the normal distribution of the ciphertext while concealing its real value. The method operates as follows: Given a ciphertext range $[a, b]$ and a number of intervals n , the range is divided into n equal-length disjoint intervals $I_i = [a + (i - 1)\Delta, a + i\Delta)$, where $\Delta = \frac{b-a}{n}$ and $i = 1, 2, \dots, n$. For each ciphertext x , it is mapped to a random variable $y \sim N(\mu_i, \sigma^2)$, where $\mu_i = a + (i - 0.5)\Delta$ represents the midpoint of the interval I_i that contains x , and σ^2 represents a chosen variance. This approach ensures the mapped data adheres to a normal distribution and prevents data mixing from different intervals. Furthermore, this method enables encoding multiple bits of information in a single interval by subdividing it into 2^k sub-intervals, each corresponding to a k -bit binary number. Subsequently, the sub-interval that matches the binary representation of x is selected, and y is generated within it. This approach can enhance the mapping efficiency and security, assuming the interval length is sufficiently large.

To decode, the interval center closest to each mapped value is located, and its original interval is identified to recover the ciphertext value. Additionally, the precision of this location process is crucial for accurate decoding. The mapped value, denoted as y , determines its interval number $i = \lfloor \frac{y-a}{\Delta} \rfloor + 1$, where $\lfloor \cdot \rfloor$ denotes the floor function. For intervals encoding multiple bits, the sub-interval is identified based on y and μ_i to derive the binary ciphertext. This is then converted into a 01 sequence to obtain the decoding result. This method remains straightforward to decode yet challenging to breach, as attackers are unable to access the interval partition and normal distribution parameters. Algorithms 3 and 4 illustrate the whole process.

Utilizing the ND method enables easy achievement of high-capacity message embedding through the adjustment of the number of bits per interval. This approach surpasses the capacity limitations of traditional generative steganography, achieving a capacity that exceeds 3.0 bpp.

It's important to clarify that ND is an encoding method that distributes bits across a Gaussian noise tensor based on predefined statistical properties. It does not provide cryptographic security. Similar to RS, it focuses on enabling the reversible transformation of data for the diffusion model. The security of the overall CGIS system does not rely on ND's encoding mechanism but on the generative process of creating stego images.

3.2.3 Sensitivity to the mean and variance of noise

The investigation focused on the effect of the original noise distribution on the conditional diffusion model. It was found that the model produces optimal denoising results when the noise adheres to the standard normal distribution, as

Algorithm 4 Noise to binary of ND.

Input: Ciphertext c , value range of ciphertext $[a, b]$, number of intervals n , number of bits per interval k , variance of normal distribution σ^2

Output: Plaintext m

```

1: Compute the length of each interval  $\Delta = \frac{b-a}{n}$ 
2: Divide  $c$  into segments  $y_1, y_2, \dots, y_t$ 
3: for  $i = 1$  to  $t$  do
4:   Compute the index of the interval that contains  $y_i$  as  $j_i = \lfloor \frac{y_i - a}{\Delta} \rfloor + 1$ 
5:   Compute the sub-interval of the  $j_i$ th interval that contains  $y_i$  as  $[a + (j_i - 1)\Delta + (l_i - 1)\frac{\Delta}{2^k}, a + (j_i - 1)\Delta + l_i\frac{\Delta}{2^k}]$ , where  $l_i = 1, 2, \dots, 2^k$ 
6:   Convert  $l_i$  to binary representation  $b_{l_i}$ 
7:   Pad  $b_{l_i}$  with leading zeros to make it  $k$ -bit long
8: end for
9: Concatenate  $b_{l_1}, b_{l_2}, \dots, b_{l_t}$  as the binary representation of the plaintext  $b_m$ 
10: Convert  $b_m$  to decimal representation  $m$ 
11: return  $m$ 

```

illustrated in Fig. 2. To preserve this distribution, two ciphertext noise bi-directional mapping methods were proposed to enable reversible and efficient mapping. This feature is crucial for diffusion model steganography.

Here, we clarify how noise mapping enables coverless embedding. In the RS method, the binary ciphertext is transformed into Gaussian-like noise. This noise, carrying the secret information, becomes an input for the conditional diffusion model. Instead of modifying an existing cover image,

the conditional diffusion model uses this noise and semantic prompts related to the image's application scenario to generate a new container image. For example, if the application scenario is a portrait image, the semantic prompt could be "portrait of a person". The model then synthesizes a new image from scratch, with the secret information already integrated within the noise. Similarly, in the ND method, the ciphertext is mapped to a more complex Gaussian-distributed noise. This noise is also fed into the conditional diffusion model. By carefully designing the interval partitioning and normal distribution parameters in the ND method, more secret information can be packed into the noise. The conditional diffusion model then generates a container image that appears natural and contains no trace of modification to any pre-existing media. At the receiving end, the inverse mapping process is used to extract the secret information from the noise within the container image. This process ensures that the secret information is securely transmitted without relying on or modifying pre-existing media, achieving true coverless embedding.

3.3 Gaussian mixture model-based image denoiser

In scenarios involving real-world transmission and steganography analysis attacks, a prevalent issue exists within existing steganography methods based on the diffusion model. These methods evenly distribute the ciphertext across the container

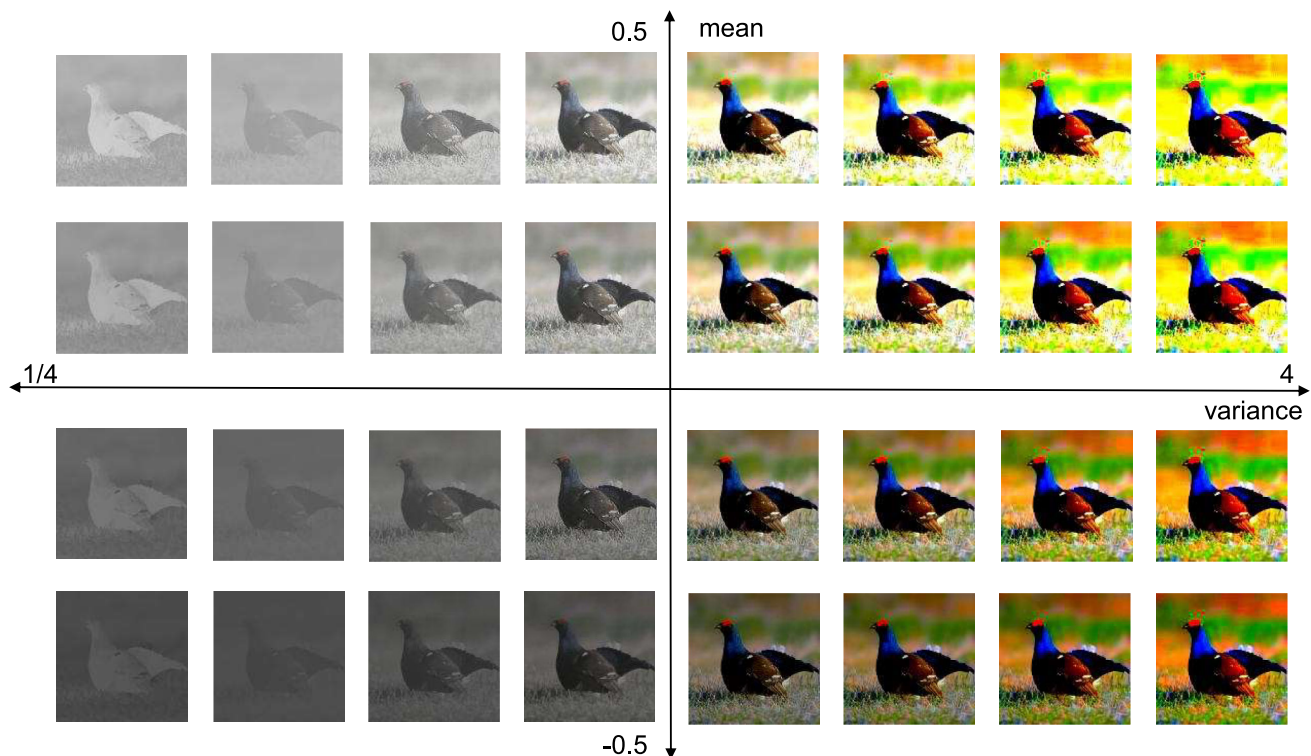


Fig. 2 The generated images are affected by the mean and variance of stego noise

image, leading to distortion during transmission that impacts the image's clarity and pixel values, ultimately diminishing decoding accuracy. To address this issue, an image denoising method is proposed, grounded in the Gaussian Mixture Model (Yuan et al. 2020).

The noise generated in Section 4.2 via bidirectional mapping (RS or ND) serves as the input to the diffusion model in Section 4.4, embedding the secret message into the generated image. However, real-world distortions during transmission (e.g., noise, compression) can degrade this image, necessitating the GMM-ID denoiser described here to restore it prior to extraction. We posit that both the noise addition by the diffusion model and the noise attacks during transmission can be modeled through a Markov random process, thus leading to the development of a dynamic model to simulate the image's distortion and recovery processes. Kernel functions are utilized to assess the local similarity of images, aiming to preserve their edge and texture features. Ultimately, the iterative convergence algorithm inherent in the Gaussian Mixture Model is employed to facilitate image denoising and numerical restoration. This process is delineated as follows:

Step 1. Extract the matrices corresponding to the three channels of the image for use in subsequent denoising and restoration operations.

Step 2. Normalize the color values for each pixel to the interval $[0, 1]$, serving as the initial state vector x_0 .

Step 3. Define the dynamic model of the image as a Markov random field as follows:

$$p(x_k | x_{k-1}) = \frac{1}{Z} \exp\left(-\sum_{i=1}^m \sum_{j=1}^n \phi(x_{ij}^k, x_{ij}^{k-1}) - \sum_{i=1}^m \sum_{j=1}^n \sum_{s \in N(i, j)} \psi(x_{ij}^k, x_{is}^k)\right) \quad (2)$$

where Z represents a normalization constant, and $\phi(x_{ij}^k, x_{ij}^{k-1})$ represents a unary potential function describing the change in color values of each pixel, defined by a normal distribution:

$$\phi(x_{ij}^k, x_{ij}^{k-1}) = \frac{1}{2Q} (x_{ij}^k - x_{ij}^{k-1})^2 \quad (3)$$

where Q is a positive constant indicating the smoothness of the image. $\psi(x_{ij}^k, x_{is}^k)$ is characterized as a binary potential function describing the similarity in color values between adjacent pixels, defined as an exponential function:

$$\psi(x_{ij}^k, x_{is}^k) = \gamma \|x_{ij}^k - x_{is}^k\|^2 \quad (4)$$

where γ is a positive constant representing the weight of similarity among adjacent pixels, $N(i, j)$

defines a neighborhood set containing the coordinates of the four adjacent pixels for the pixel at column j and row i , e.g., $N(2, 3) = \{(1, 3), (2, 2), (2, 4), (3, 3)\}$. Through this approach, the dynamic model more effectively delineates the conditional probability distribution of color values for each pixel, thus enabling more precise image predictions.

Step 4. Define the observation model of the image as a mixture of Gaussian distributions:

$$p(z_{ij}^k | x_{ij}^k) = \pi N(z_{ij}^k | h(x_{ij}^k), R_k) + (1 - \pi) N(z_{ij}^k | \mu, \sigma^2) \quad (5)$$

where π represents the mixing coefficient, reflecting the probability that the observed data for each pixel is generated by the kernel function. $N(z_{ij}^k | h(x_{ij}^k), R_k)$ denotes a normal distribution corresponding to the probability density function of each pixel generated by the kernel function, and $h(x_{ij}^k)$ denotes a kernel function used to measure the image's local similarity, for instance:

$$h(x_{ij}^k) = \exp(-\gamma \|x_{ij}^k - x_r\|^2) \quad (6)$$

where γ is a positive kernel parameter, and x_r is a fixed reference vector, such as all 1s or all 0s. $\|\cdot\|$ represents the Frobenius norm of the matrix. Thus, when the image's color value is close to x_r , $h(x_{ij}^k)$ tends towards 1; conversely, when it is far from x_r , $h(x_{ij}^k)$ approaches 0. This observation model can effectively describe the edge and texture features of an image. $N(z_{ij}^k | \mu, \sigma^2)$ denotes a normal distribution that constitutes the probability density function for each pixel influenced by noise or outliers, with μ and σ^2 representing the mean and variance of such disturbances, which can be estimated through the image's global statistics. Consequently, our observation model more precisely characterizes the probability distribution of the observed data for each pixel, thereby facilitating a more accurate depiction of the image.

Step 5. Define the covariance matrix for the image's observed noise as an adaptive covariance matrix:

$$R_k = \alpha \log(1 + \beta S_k) + \delta C_k \quad (7)$$

where α , β , and δ represent positive constants, and S_k denotes the image's information entropy, defined as follows:

$$S_k = -\sum_{i=1}^m \sum_{j=1}^n p_{ij} \log p_{ij} \quad (8)$$

where m and n represent the number of rows and columns in the image, respectively, and p_{ij} denotes the probability density function value for pixels in column j and row i . This can be estimated using a histogram. Consequently, when the image's information entropy is high, indicating greater uncertainty, R_k increases, leading to a reduced Kalman gain and a lower degree of trust in the observed data. Conversely, a low information entropy signifies less uncertainty, resulting in a higher Kalman gain and greater trust in the observed data. C_k , the covariance matrix for each pixel's observed noise, can be estimated with the following formula:

$$C_k = \frac{1}{|N(i, j)|} \sum_{s \in N(i, j)} (z_{ij}^k - z_{is}^k)^2 \quad (9)$$

where $N(i, j)$ constitutes a neighborhood set comprising the coordinates of the four adjacent pixels around the pixel at column j and row i . Thus, our observational noise covariance matrix can utilize the spatial information of the image more effectively, yielding a more precise update of the image.

Step 6. Define the initial state covariance matrix of the image as an identity matrix:

$$P_0 = I \quad (10)$$

where I denotes an identity matrix matching the image's dimensions, representing the image's initial uncertainty.

Step 7. Use a Gaussian Kalman-like filter to iteratively update the state and covariance matrix of the image until convergence is achieved or the maximum number of iterations is reached, defined as follows:

$$K_k = P_k^- H^T (H P_k^- H^T + R_k)^{-1} \quad (11)$$

$$x_k = x_k^- + K_k(z_k - h(x_k^-)) \quad (12)$$

$$P_k = (I - K_k H) P_k^- \quad (13)$$

where P_k^- represents the prediction covariance matrix, x_k^- signifies the predicted state, and H is identified as the Jacobian matrix of the observational model, given by the following formula:

$$H = \left. \frac{\partial h}{\partial x} \right|_{x=x_k^-} = -2\gamma(x_k^- - x_r)h(x_k^-) \quad (14)$$

Step 8. Multiply the updated state vector x_k by 255 and convert it to an integer to obtain the color value for the denoised and restored image.

The denoised image is then processed by the reverse diffusion steps in Section 4.2 to recover the original

noise, from which the secret message is extracted. This shows the crucial role of the GMM-ID denoiser in the entire steganography process, connecting the image generation and noise-embedding steps in Sections 4.2 and 4.4 with the final message extraction.

3.4 Conditional diffusion model

To integrate the secret information into the generation process of Stable Diffusion, we strictly define the embedding location of the noise tensor. Unlike traditional methods that add perturbations to an existing image, our framework utilizes the mapped noise tensor N (carrying the secret message) to replace the initial random latent variable z_T at the start of the reverse diffusion process. Specifically, in the latent space of the VAE, we set $z_T = N$. This ensures that the steganographic signal serves as the foundational "seed" for generation. The DDIM sampler then iteratively denoises this specific latent z_T from timestep T to 0, transforming the secret-laden noise into a coherent image structure.

Furthermore, DDIM enhances Denoising Diffusion Probabilistic Models (DDPM) by utilizing a non-Markovian diffusion process, maintaining the same training objective while enabling significantly faster sampling. Its key innovation lies in using only the initial noise, rather than introducing new noise at each step. This approach eliminates the need for computing noise variance and gradients, streamlining the process and further improving efficiency.

The following equation describes the noise addition process in stable diffusion:

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I) \quad (15)$$

The denoising process in stable diffusion is articulated through the following equation:

$$x_t = x_0 + (1 - \alpha_t)(x_{t-1} - x_0) - \beta_t \nabla_x \log p_\theta(x_{t-1}|x_0) \quad (16)$$

where x_t represents the denoised data at step t , x_0 denotes the initial random noise, and β_t indicate the diffusion coefficients at step t , and p_θ embodies the denoising network, modelable by a neural network.

Utilizing DDIM enables the achievement of reversible noise addition and denoising processes in deterministic diffusion models, significantly enhancing information extraction accuracy. Furthermore, in comparison to general conditional diffusion models, it possesses the capability to process more complex and extensive prompts, rendering it applicable across a broader spectrum of application scenarios.

4 Experiment

To validate the efficacy of the CGIS framework, a series of experiments were conducted and compared against existing generative steganography models. These experiments primarily investigated the impact of diverse mapping methods on experimental performance, as well as the performance of various generative models in both identical and distinct experimental scenarios. Furthermore, this research incorporated the influence of image transmission format on transmission performance.

4.1 Implement detail

In the CGIS framework, the conditional diffusion model utilizes pre-trained Stable Diffusion. For performance comparison with existing generative steganography models, the ILSVRC 2012 ImageNet dataset (Russakovsky et al. 2015) is employed as the experimental dataset.

The ILSVRC 2012 ImageNet has training, validation, and test data. Based on WordNet, it has 1000 non-overlapping object categories (low-level synsets) as part of the larger ImageNet, with 860 high-level synsets as parent classes. Information is in data/meta.mat. Training images are in JPEG format in synset-named tar files, totaling 1,281,167 with 732–1300 per synset. There are 50,000 validation images with 50 per synset and ground truth. 100,000 test images have no public ground truth. At least 100 training and all validation and test images have bounding box annotations in PASCAL VOC format with XML files. It's widely used for visual tasks like image classification, object detection, and fine-grained classification and crucial for deep learning model development and evaluation.

From the ILSVRC 2012 ImageNet dataset, 50 classes are selected, with 400 images randomly chosen from each class for the experimental dataset. Subsequently, the dataset is divided into training, validation, and test sets in an 8:1:1 ratio. All subsequent experiments are conducted on this divided dataset. Furthermore, all images are resized to 128×128 for input into the model for training.

Regarding the conditional inputs, the semantic prompt P is first encoded by the pre-trained CLIP text encoder to produce high-dimensional text embeddings. These embeddings are injected into the U-Net backbone via the **Cross-Attention mechanism**. Mathematically, the interaction is modeled as $\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d}})V$, where the query Q is derived from the intermediate spatial features of the noisy latent z_t , while the keys K and values V are projected from the prompt embeddings. This mechanism ensures that while the initial noise z_T determines the hidden steganographic content, the semantic prompt continuously guides the

denoising trajectory to ensure the visual appearance aligns with the user's description.

4.2 Hyperparametric study of GMM-ID

In the GMM-ID method, several parameters requiring tuning influence image denoising performance. Parameters are classified into internal, related to the dynamical and observation models, and external, derived from the noisy image. The most impactful internal parameters on the denoising results include:

- Q , controlling image smoothness via the unary potential function of the dynamical model. A higher Q value implies a smoother image with reduced pixel color variation.
- γ , determining adjacent pixel similarity weight via the binary potential function of the dynamical model and the observation model's kernel function. A larger γ value implies higher neighboring pixel similarity, with more minor color value differences. The ranges for grid search on Q and γ were selected based on the statistical properties of natural image gradients and local texture consistency, ensuring that the optimization covers the most relevant parameter space for preserving edge details.
- π , representing the probability that observed pixel data is generated by a kernel function, affecting the observation model's Gaussian mixture distributions. A higher π value implies a greater likelihood of pixel data being generated by the kernel function, enhancing image edge and texture features. Experimental findings indicated minimal effect from changing π 's value, thus setting π to 0.5.

The external observation parameters are set as follows:

- μ and σ^2 , representing the mean and variance of noise or outliers, influence the observation model's mixed Gaussian distribution. Lower values of μ and σ^2 reduce noise or outlier impact on observations, bringing the observed image closer to the true image. These parameters can be estimated from the image's statistical properties. Assuming a moderate noise level in transmission, μ is set to 0.25 and σ^2 to 0.5.
- α , β , and δ , as adaptive parameters for the observation noise covariance matrix, determine its size and shape. Higher values of α , β , and δ enlarge the observation noise covariance matrix, reducing trust in observations relative to state predictions. Empirical evidence from extensive local experiments sets α at 0.95, β at 0.9, and δ at 0.7.

To determine the optimal values of Q and γ , GMM-ID was used to denoise and repair the container image across embedding rates from 1.0 bits per pixel (bpp) to 6.0 bpp. A complete image steganography and decoding process was simulated, recording the bit error rate (BER), detection error (Pe), and image quality under various GMM-ID parameters. BER, defined as the fraction of bits differing between the original and extracted ciphertext, is given by:

$$BER = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(x_i \neq y_i) \quad (17)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Lower BER values indicate improved information transmission. Pe is an indicator of the performance of classifier models. Pe is defined as follows:

$$Pe = \min_{P_{FA}} \frac{1}{2} (P_{FA} + P_{MD}) \quad (18)$$

where P_{FA} and P_{MD} are the rates of false-alarm and miss-detection errors. A Pe value of 0.5 means that the classifier can not distinguish two classes completely. It reflects SiaStegaNet's (You et al. 2021) ability to classify images as containing steganographic information or not. Image quality, measured by the Fréchet Inception Distance (FID) (Heusel et al. 2017), reflects the visual similarity between the original and container images. Lower FID values signify reduced distortion. Experimental results for various parameters are illustrated in Fig. 3.

Figure 4 showcases denoised images (128×128) with specific (Q, γ) values. These images primarily display the detailed effects of noise generated through network transmission or steganalysis attacks. It is observed that inappropriate parameter values introduce additional random noise to the images during the denoising process. Simultaneously, fine-tuning these parameters can achieve optimal effects for various transmission tasks.

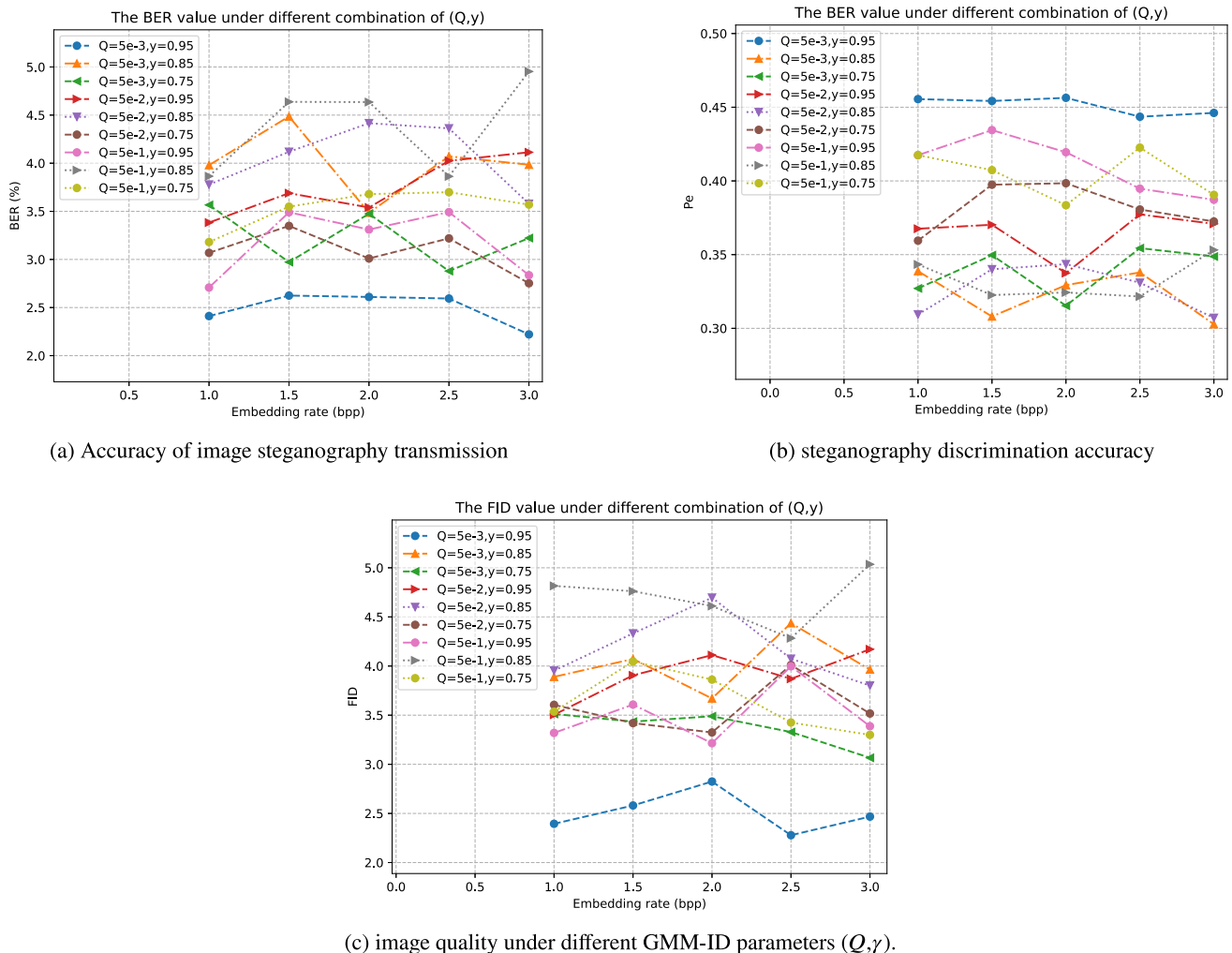


Fig. 3 (a) Accuracy of image steganography transmission (b) steganography discrimination accuracy (c) image quality under different GMM-ID parameters (Q, γ)

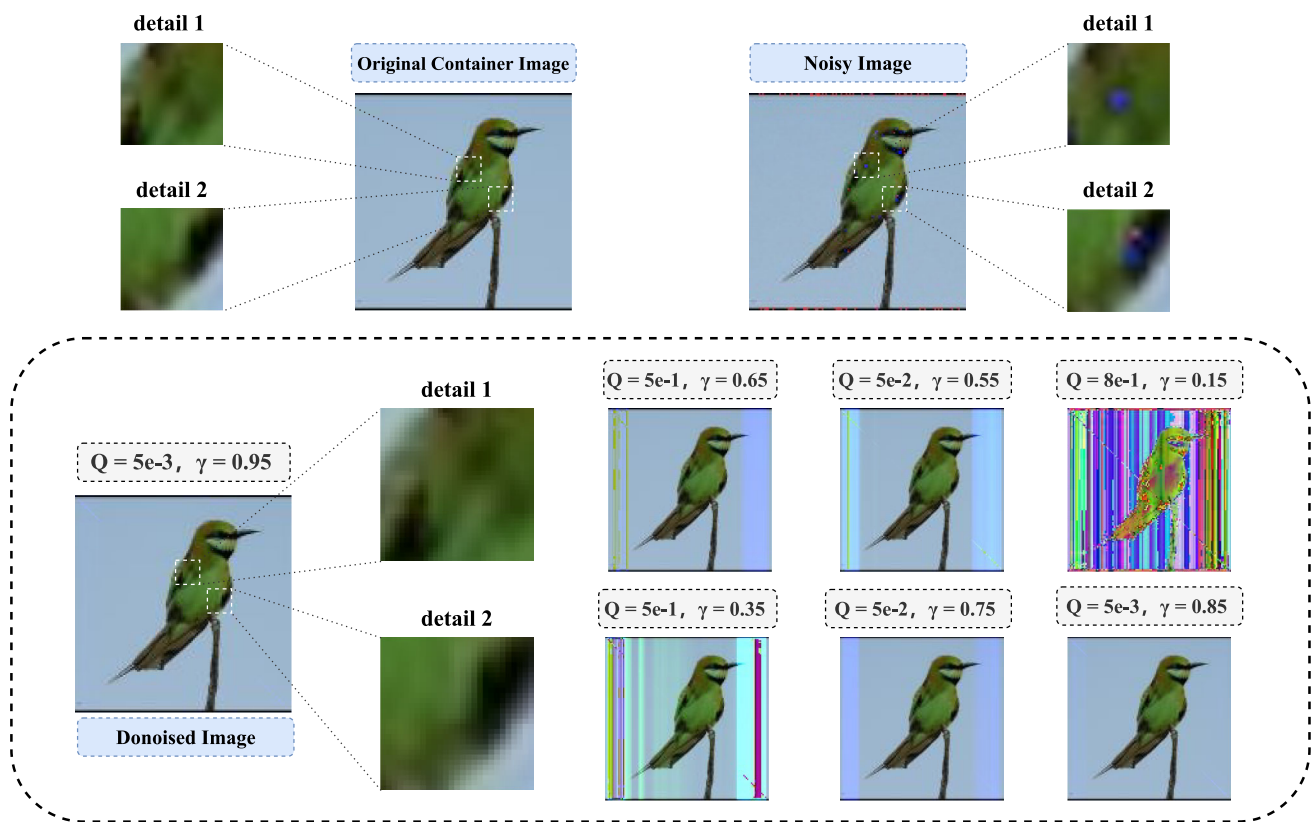


Fig. 4 The demo of GMM-ID's image denoising effect under different parameters

4.3 Aboliation study

4.3.1 Time complexity analysis of GMM-ID

To quantitatively assess the computational overhead introduced by the GMM-ID module, we recorded the average execution time for the secret message extraction phase. The experiments were conducted on a computing platform equipped with an NVIDIA RTX 4090D GPU. The baseline extraction process (without denoising) requires approximately 0.043 seconds per image. Enabling the GMM-ID with the default iteration count $K = 5$ increases the total extraction time to roughly 0.197 seconds per image. While this represents an additional computational cost of 0.15 seconds, the complexity remains linear $\mathcal{O}(K \cdot N)$ with respect to the image size N and iterations K . Crucially, this marginal increase in

latency yields a substantial gain in robustness, effectively mitigating high-intensity noise and compression artifacts that would otherwise render the secret message unreadable. For most practical steganographic applications where transmission reliability takes precedence over millisecond-level real-time performance, this trade-off is fully justified.

4.3.2 Evaluation of different combination strategies

This study compares the performance of two mapping methods (RS, ND) in the steganography process of the diffusion model, both with and without GMM-ID, on the same dataset. Three metrics are employed to evaluate their performance: BER, Pe, and FID. The embedding rate is varied from 1.0 to 6.0 bpp, generating the transmission ciphertext as a random binary sequence matching the length product of the embed-

Table 1 Aboliation study on two mapping methods and denoise method

Mapping	1.0 bpp			3.0 bpp			6.0 bpp		
	BER (%)	Pe	FID	BER	Pe	FID	BER (%)	Pe	FID
RS (w/ GMM-ID)	3.89	0.45	2.46	4.91	0.45	2.37	5.13	0.45	2.51
RS (w/o GMM-ID)	6.98	0.40	3.21	6.49	0.39	3.07	6.42	0.40	3.14
ND (w/ GMM-ID)	2.74	0.48	2.68	2.57	0.48	2.59	2.25	0.48	2.36
ND (w/o GMM-ID)	5.05	0.45	2.84	5.77	0.45	2.79	5.48	0.45	2.98

ding rate and image size. For instance, a 128×128 image at 1.0 bpp embedding rate results in a ciphertext length of $128 \times 128 = 16384$ bits. Results are presented in Table 1.

Notably, the data in Table 1 exhibits a counter-intuitive phenomenon where the Bit Error Rate (BER) decreases as the amount of embedded secret information increases. This trend contrasts with traditional cover-modification steganography, where higher payloads typically introduce greater distortion and extraction difficulty. This phenomenon is attributed to the generative nature of the proposed CGIS framework. The core diffusion model is pre-trained to map standard Gaussian noise $\mathcal{N}(0, \mathbf{I})$ to the natural image manifold. At lower embedding rates, the mapped latent variables derived from sparse secret message bits may statistically deviate slightly from the ideal Gaussian prior, resulting in a “distribution mismatch” that causes minor drifts during the reverse diffusion process. Conversely, at higher embedding rates, the embedded payload consists of dense, pseudo-random encrypted binary sequences. According to the Law of Large Numbers, these dense sequences, when mapped to the latent space, statistically approximate Gaussian white noise more perfectly. This improved “distribution matching” enhances the stability of the DDIM generation trajectory, thereby reducing quantization errors and lowering the BER in high-capacity scenarios.

In the practical implementation and analysis, it was discovered that the RS method resamples noise at fixed intervals. This particular approach plays a significant role in balancing the data distribution and maintaining the similarity of the data. However, when faced with image distortion that stems from significant noise attacks, this method imposes a requirement for precise control over the selection of the sampling threshold. If the thresholds are either too large or too small, it can cause the mapped Gaussian noise distribution to deviate from the normal distribution, thereby affecting the overall performance. Consequently, based on the RS method, the ND method is proposed. This new approach enhances the accuracy and robustness of bidirectional mapping by mapping multiple discrete intervals. Extensive experiments have

been conducted, and the results clearly demonstrate the superior efficiency of the ND method. Simultaneously, taking into consideration the potential image distortion that may occur due to network transmission or attacks from steganography analysts, the GMM-ID method is proposed. This method focuses on image denoising and restoration by modeling the Markov stochastic process dynamics system. Through a series of experiments, it has been demonstrated that GMM-ID can significantly restore image clarity. This restoration capability proves to be highly effective in resisting real-world noise attacks and ultimately enhances the overall transmission accuracy of the system.

4.4 Comparison study

The performance of the CGIS framework is evaluated by comparison with three existing generative steganography models: GSN (Wei et al. 2022), Baseline A (Xiaoxiao et al. 2024), and StegaDDPM (Peng et al. 2023). These models were trained on the same dataset, noting that traditional generative models have limited embedding rates due to algorithmic constraints. Consequently, embedding rates were set according to the capabilities of the GSN, StegaDDPM and Baseline A models, aiming to approximate their maximum embedding capacities. Especially for the Baseline A model, we selected the steganographic capacity recommended in the original paper for comparative experiments to achieve its best performance. For CGIS, embedding rates of 1.0bpp, 3.0bpp, and 6.0bpp were selected. Two real-world transmission scenarios were also simulated: network transmission and JPEG compression. In the network transmission scenario, Gaussian noise with variances (5, 10, 20) and uniformly distributed noise with ranges (5, 10, 20) were added to the transmitted images. In the JPEG compression scenario, images were generated and saved in JPEG format using Stable Diffusion, with quality factors (QF = 90, 70, 50). Experimental results are presented in Tables 2, 3, and 4.

Table 2 Comparison of differences between baseline models under different noise (Gaussian distributed) conditions

Model	Embedding rate (bpp)	$\sigma^2 = 5$			$\sigma^2 = 10$			$\sigma^2 = 20$		
		BER (%)	Pe	FID	BER (%)	Pe	FID	BER (%)	Pe	FID
GSN (Wei et al. 2022)	3.44e-3	6.93	0.38	10.68	14.03	0.32	17.94	20.14	0.28	25.48
	1.36e-2	9.75	0.34	12.83	16.58	0.29	22.13	24.44	0.24	27.35
StegaDDPM (Peng et al. 2023)	1.0	7.92	0.40	7.94	11.14	0.36	13.95	21.11	0.32	16.90
	3.0	8.57	0.37	8.51	14.11	0.33	14.77	23.64	0.30	18.13
Baseline A (Xiaoxiao et al. 2024)	1.25e-2	6.87	0.42	3.99	13.19	0.40	5.00	18.16	0.37	4.95
	6.25e-2	7.52	0.41	3.46	13.16	0.40	4.82	19.69	0.37	5.18
	1.0	8.26	0.42	4.44	14.46	0.40	5.98	20.95	0.38	6.14
CGIS	1.0	3.85	0.46	2.84	4.31	0.45	3.06	5.46	0.43	3.28
	3.0	3.35	0.46	2.91	4.31	0.45	3.03	5.82	0.43	3.21
	6.0	3.70	0.46	2.90	4.77	0.44	3.15	5.24	0.43	3.25

Table 3 Comparison of differences between baseline models under different noise (uniformly distributed) conditions

Model	Embedding rate (bpp)	noise range = 5			noise range = 10			noise range = 20		
		BER (%)	Pe	FID	BER (%)	Pe	FID	BER (%)	Pe	FID
GSN (Wei et al. 2022)	3.44e-3	7.12	0.37	11.02	14.27	0.32	18.18	24.29	0.27	25.82
	1.36e-2	10.04	0.33	13.17	16.82	0.28	20.18	25.69	0.23	28.19
StegaDDPM (Peng et al. 2023)	1.0	8.21	0.38	8.26	11.43	0.35	14.24	21.40	0.32	17.19
	3.0	10.86	0.36	10.83	14.40	0.32	17.06	24.03	0.30	20.42
Baseline A (Xiaoxiao et al. 2024)	1.25e-2	7.55	0.42	4.19	13.49	0.40	5.20	19.11	0.37	5.15
	6.25e-2	7.81	0.42	3.66	13.74	0.39	4.92	19.34	0.37	5.38
	1.0	9.25	0.42	4.54	15.68	0.39	5.68	20.60	0.37	6.24
CGIS	1.0	3.14	0.46	3.02	4.60	0.44	2.56	5.75	0.42	3.43
	3.0	3.64	0.46	2.99	4.30	0.44	3.29	5.11	0.42	3.40
	6.0	3.79	0.46	2.86	4.26	0.44	3.15	5.53	0.42	3.47

The CGIS framework outperforms other models across all three metrics at various noise levels, balancing delivery accuracy, security, and image quality under different JPEG compression factors. With increasing steganographic capacity, traditional generative models and StegaDDPM exhibit reduced robustness, leading to higher and less stable message delivery error rates. CGIS achieves secure, stable, and efficient message transmission through a powerful message mapping mechanism and the GMM-ID filter, demonstrating strong practicability. The effects of different models under network transmission and noise attack are illustrated in Figs. 5 and 6, respectively.

To clearly present the comparison results with traditional coverless steganography methods, we conducted a detailed analysis of the data in Tables 2, 3, and 4. Under Gaussian noise attack with a variance of 5 and an embedding rate of 1.0 bpp, CGIS achieves a bit error rate (BER) of 3.85%, significantly lower than GSN (6.93%) and StegaDDPM (7.92%). This indicates that CGIS demonstrates higher transmission accuracy, with a noticeably lower error rate in information

extraction. Furthermore, as the noise variance increases to 10 and 20, the BER of CGIS remains relatively stable compared to the other two models. For instance, at a variance of 20, CGIS maintains a BER of 5.46% at 1.0 bpp, whereas GSN surges to 17.94% and StegaDDPM reaches 14.77%, highlighting CGIS's superior robustness against noise interference.

In terms of image quality, as measured by the Fréchet Inception Distance (FID), CGIS outperforms its counterparts significantly. At an embedding rate of 3.0 bpp under normal circumstances (without additional noise in the network transmission), CGIS achieves an FID of 2.91, whereas GSN and StegaDDPM exhibit substantially higher values of 16.58 and 14.11, respectively. Since lower FID values indicate less distortion, these results confirm that CGIS preserves the visual quality of the generated images more effectively. Even under JPEG compression with varying quality factors, CGIS continues to demonstrate exceptional performance. For example, when the JPEG quality factor is 50 at an embedding rate of 3.0 bpp, CGIS achieves an FID of 4.81, whereas GSN and Ste-

Table 4 Comparison of baseline models under different JPEG compression quality

Model	Embedding rate (bpp)	QF = 90%			QF = 70%			QF = 50%		
		BER (%)	Pe	FID	BER (%)	Pe	FID	BER (%)	Pe	FID
GSN (Wei et al. 2022)	3.44e-3	9.22	0.37	12.14	14.70	0.33	19.49	24.83	0.26	24.29
	1.36e-2	11.16	0.33	14.80	18.06	0.28	22.13	27.70	0.22	27.56
StegaDDPM (Peng et al. 2023)	1.0	7.44	0.37	8.21	13.53	0.34	14.86	20.08	0.32	20.37
	3.0	8.26	0.34	9.72	15.72	0.31	16.02	23.28	0.28	23.61
Baseline A (Xiaoxiao et al. 2024)	1.25e-2	6.31	0.42	3.67	12.28	0.40	4.17	18.03	0.37	5.22
	6.25e-2	7.06	0.42	4.11	11.62	0.40	4.38	19.23	0.38	5.21
	1.0	8.16	0.42	4.74	13.94	0.39	5.84	21.59	0.37	5.92
CGIS	1.0	3.18	0.45	2.64	5.31	0.44	3.54	6.48	0.42	4.69
	3.0	3.09	0.45	2.82	5.64	0.44	3.33	7.00	0.42	4.81
	6.0	2.88	0.45	2.87	5.89	0.44	3.57	7.15	0.42	4.79

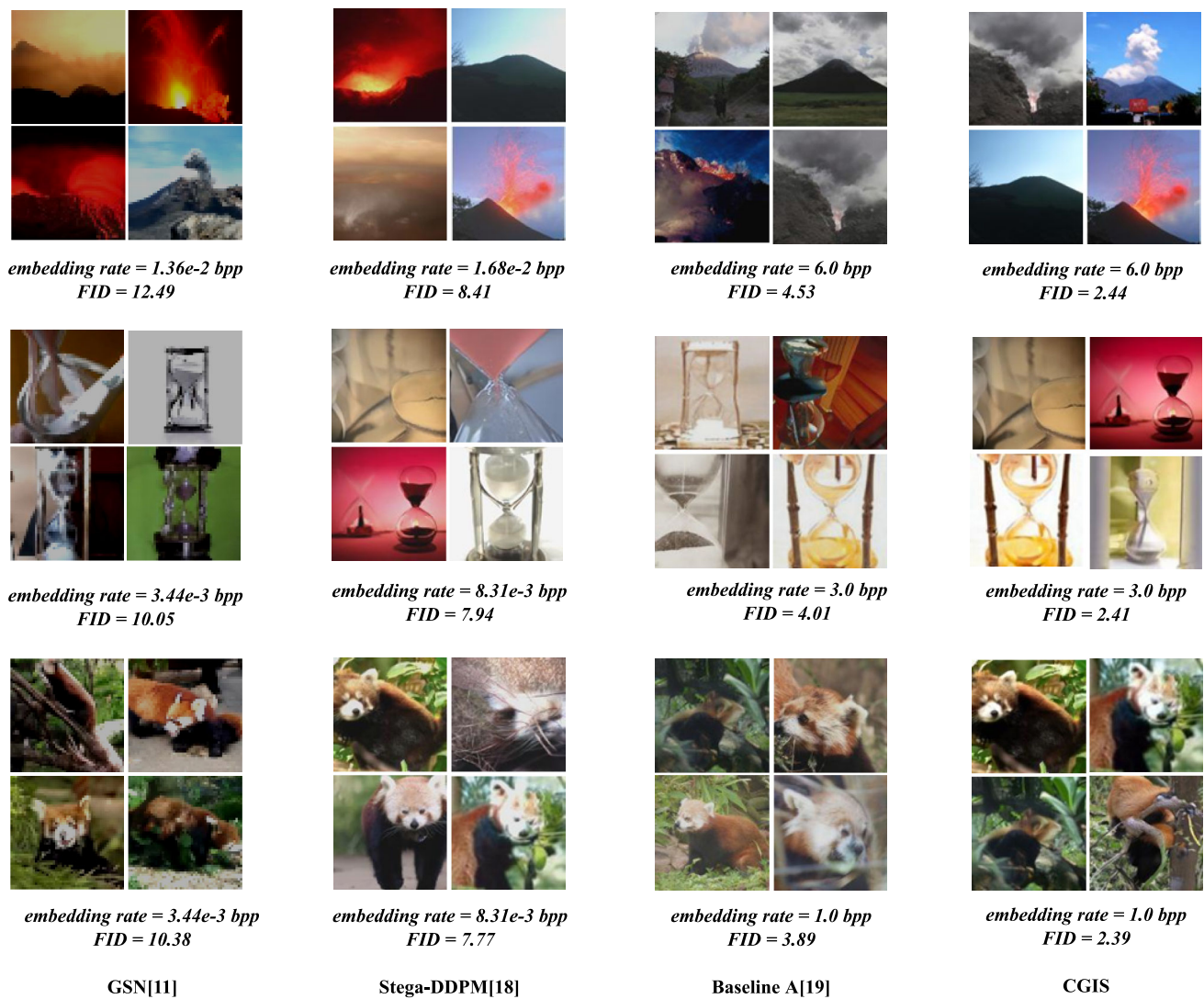


Fig. 5 Images generated by baseline models at different embedding rates and prompts

gaDDPM deteriorate significantly, with FID values of 24.29 and 23.61, respectively. These findings further underscore CGIS's resilience to compression artifacts, making it a more reliable solution for real-world image steganography applications.

4.5 Security Analysis

Steganographic security is a critical metric for evaluating an algorithm's resilience against steganalytic attacks. In this work, we analyze the anti-detection performance of the proposed CGIS method using two state-of-the-art detectors: StegExpose (Olson et al. 2017) (statistical analyzer) and SRNet (Boroumand et al. 2018) (deep learning-based detector). Comparative experiments are performed against adaptive LSB-HILL, StegaDDPM, and Baseline A, employ-

ing 10,000 stego images generated from the LAION-10K (Schuhmann et al. 2022) and COCO (Lin et al. 2014) datasets at an embedding rate of 1.0 bpp. Key evaluation metrics include:

StegExpose probability of error (PE) Quantifies the steganalysis tool's error rate in distinguishing between steganographic images and non-steganographic generated samples (benign samples). It is calculated as $\min_{P_{FA}} \frac{1}{2}(P_{FA} + P_{MD})$, where P_{FA} and P_{MD} denote the false alarm and missed detection rates, respectively. An ideal P_E of 50% indicates that the detector cannot differentiate the stego images from benign ones better than random guessing, signifying high security.

SRNet classification accuracy Reflects the detector's ability to distinguish stego images from benign samples (lower accuracy correlates with stronger security).

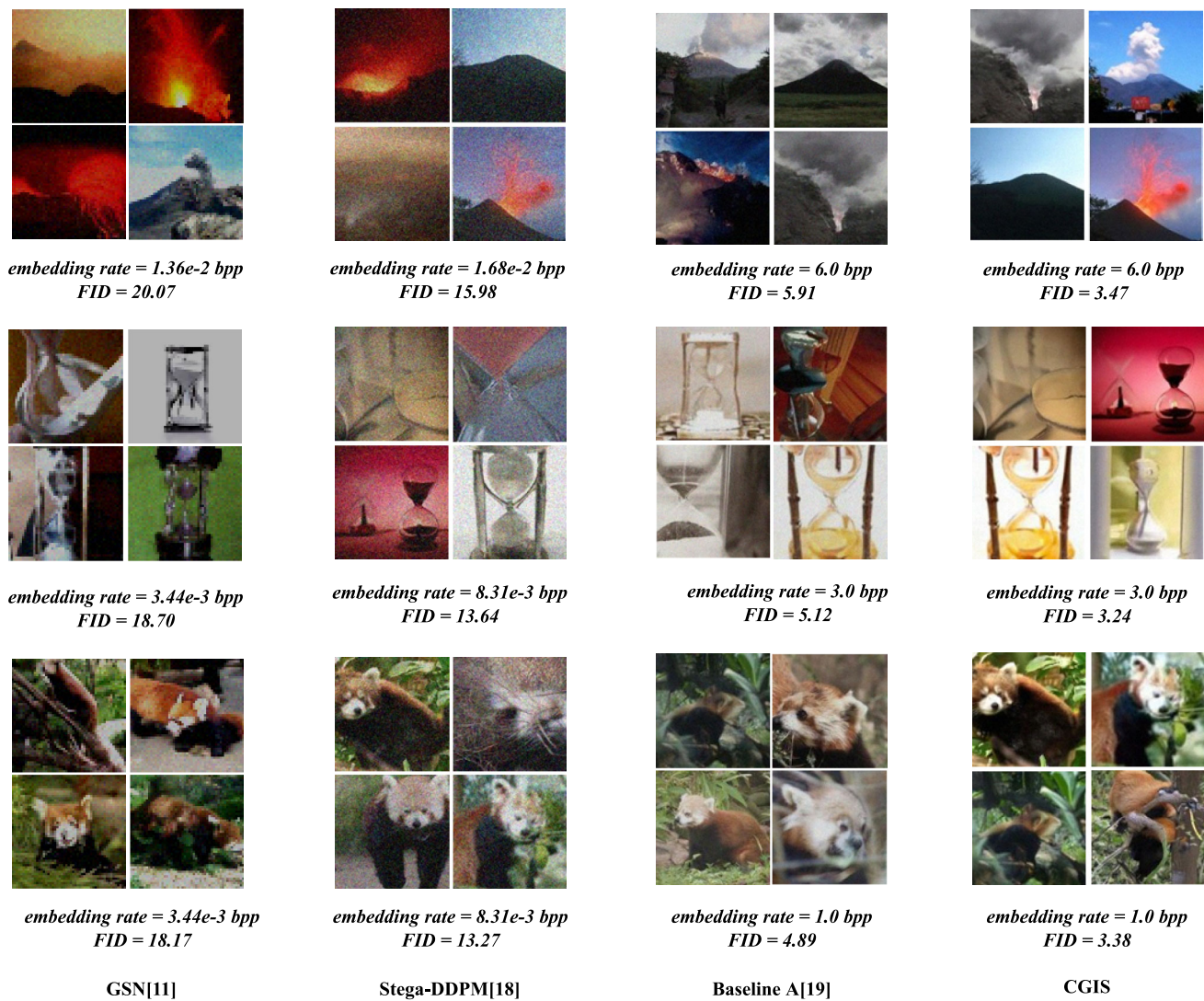


Fig. 6 Images generated by baseline models at different embedding rates and prompts against Gaussian noise attack

Bit error rate (BER) Quantifies extraction robustness under JPEG compression (QF=70) and additive Gaussian noise ($\sigma^2=20$).

As summarized in Table 5, CGIS achieves a StegExpose PE of 47.3% at 1.0 bpp—nearly matching the theoretical random-guessing threshold (50%) and surpassing StegaDDPM (34.2% PE) and traditional LSB methods (18.7% PE). This highlights CGIS’s ability to minimize statistical devi-

ations between stego and natural images, a weakness of pixel-domain approaches.

Furthermore, SRNet attains its lowest accuracy (53.9%) against CGIS-generated stego images, underscoring the diffusion model’s advantage in aligning latent distributions. In contrast, Baseline A’s task-specific generative design exhibits higher vulnerability (72.1% SRNet accuracy), revealing inherent security trade-offs in prompt-conditioned architectures.

Table 5 Security performance comparison under different embedding rates

Method	Embedding Rate (bpp)	StegExpose PE (%)	SRNet Accuracy (%)	BER (%)
LSB (Doo et al. 2019)	1.0	18.7	92.5	42.3
StegaDDPM (Peng et al. 2023)	1.0	34.2	65.8	12.7
Baseline A (Xiaoxiao et al. 2024)	1.0	29.5	72.1	8.6
CGIS	1.0	47.3	53.9	5.46

Table 6 Comparison of proposed method with SOTA method

Method	bpp	BER(%)	Pe	FID
Liu (Liu et al. 2022)	4.17e-2	5.20	0.42	15.92
GSD (Zhou et al. 2025)	1	3.28	0.44	6.84
CGIS	1	2.93	0.46	2.77

The proposed method further demonstrates superior robustness against channel distortions. Under Gaussian noise ($\sigma^2=20$), CGIS achieves a BER of 5.46% at 1.0 bpp, significantly lower than Hu et al.'s method (8.6% BER). This resilience is attributed to the GMM-ID module, which leverages noise modeling within a Markov random field framework to dynamically optimize covariance matrices—effectively preserving edge features while suppressing distortion.

In summary, CGIS strikes an optimal balance between security and robustness. Its statistical undetectability, resistance to deep learning-based steganalysis, and distortion-tolerant extraction collectively establish its superiority over both traditional and state-of-the-art steganographic techniques.

4.6 Performance comparison with state-of-art

In this section, we evaluate the superiority of CGIS compared to other state-of-the-art (SOTA) diffusion-based steganographic schemes, and the results are shown in Table 6. The methods proposed by Liu et al. (2022) and GSD (Zhou et al. 2025) listed in Table 6 are all steganographic methods that use GANs to generate content. CGIS is our proposed generative steganographic method based on the diffusion model.

To evaluate the performance of each method, we conduct comparative experiments on the LSVRC 2012 ImageNet dataset, focusing on three key metrics: bit error rate (BER), security (Pe), and image quality (FID). Due to variations in data embedding techniques, the payload capacities of different methods differ. For secret data extraction, stego images generated by each method are first saved as PNG images, from which the embedded data is then retrieved. For security assessment, we train the SiaStegaNet steganalysis network using 5,000 stego/benign samples generated by each method. The trained model is then employed to classify 1,000 stego/benign samples per technique, from which the detection error rate (Pe) is calculated. This evaluation provides a comprehensive understanding of each method's resistance to steganalysis detection, further highlighting the advantages of CGIS in maintaining both security and imperceptibility.

From the experimental results of the above three models, it can be seen that Liu's (Liu et al. 2022) model has obvious deficiencies in the embedding rate. Meanwhile, its performance results under the attack in a noise-free condition also show relatively obvious weaknesses. On the other hand,

while significantly improving the quality of the generated images, GSD has also successfully enhanced the performance of the ciphertext embedding rate, achieving a balance with accuracy. It can be seen that CGIS achieves the best performance in terms of security, extraction accuracy, and image quality, and it is the most powerful steganographic scheme so far.

4.7 Multi-scenario transmission accuracy

The CGIS framework utilizes a pre-trained conditional model to generate camouflaged images for various real-world scenarios, maintaining high information extraction accuracy. CGIS is compared with three other generative models across three representative scenarios: face, animal, and fruit, differing in image features, difficulties, information transmission needs, and security risks. Embedding rates are varied, and the models' performance is evaluated after transmitting the images via network transmission. Three metrics are used: message delivery accuracy, cdetection error, and image quality. Experimental results are presented in Table 7.

Table 7 demonstrates CGIS's superiority over the other two generative models in message delivery accuracy across all three scenarios. This indicates that CGIS ensures message integrity and accuracy, irrespective of image features and embedding rates. This is attributed to Stable Diffusion's ability to generate realistic images tailored to various scenarios and prompts. Furthermore, the bi-directional mapping guarantees reliable information transfer across diverse application scenarios. Thus, CGIS strikes a balance between message delivery accuracy, detection error, and image quality. In conclusion, CGIS proves more suitable and valuable for realistic multi-scenario applications compared to the other two generative models.

4.8 Message extraction accuracy according to different image format

Given that most commonly used image formats rely on lossy compression algorithms, this unavoidably causes a certain degree of degradation before the image is delivered. This degradation can manifest as blurriness, color distortion, or loss of fine details. Moreover, it might also lead to issues such as inconsistent contrast in some complex images. Since there is a significant variation in image quality loss among different image compression algorithms, comparative experiments were carried out for three commonly used ones: PNG, JPEG, and TIFF. These formats are widely used in different fields, and understanding their performance differences is crucial. By analyzing these results, we can make more informed decisions regarding which format to use based on specific requirements. Results are presented in Table 8.

Table 7 Comparison of baseline models under different real-world application scenarios via WeChat transmission

Model	Embedding rate (bpp)	Faces			Animals			Fruits		
		BER (%)	Pe	FID	BER (%)	Pe	FID	BER (%)	Pe	FID
GSN (Wei et al. 2022)	3.44e-3	8.42	0.40	10.42	9.19	0.39	10.55	8.56	0.39	9.15
	1.36e-2	11.62	0.35	11.80	10.16	0.35	11.71	11.51	0.35	12.06
StegaDDPM (Peng et al. 2023)	1.0	7.11	0.40	7.71	7.35	0.40	7.83	7.33	0.40	7.40
	3.0	8.96	0.36	9.02	8.42	0.36	9.04	8.25	0.36	8.44
Baseline A (Xiaoxiao et al. 2024)	1.25e-2	6.32	0.43	3.97	6.51	0.43	3.97	6.47	0.43	3.54
	6.25e-2	7.17	0.43	3.69	7.09	0.42	3.71	7.92	0.41	3.58
CGIS	1.0	7.93	0.42	4.31	8.06	0.43	4.35	8.75	0.42	4.42
	1.0	3.23	0.48	3.01	2.74	0.48	2.45	2.78	0.48	2.36
	3.0	2.71	0.48	2.48	2.99	0.47	2.67	3.00	0.47	2.62
	6.0	2.98	0.48	2.38	3.03	0.47	2.47	3.22	0.47	2.95

Experiments demonstrate that the two lossless compression formats, TIFF and PNG, exhibit strong robustness, making them more suitable for real-world applications.

4.9 Impact of image resolution

To evaluate the scalability and generalization capability of the proposed CGIS framework, we extended our performance evaluation beyond the baseline resolution of 128×128 . We conducted additional experiments on images resized to 64×64 and 256×256 pixels, maintaining a constant embedding rate of 2.0 bpp. This setup ensures that the total payload scales proportionally with the number of pixels.

The comparative results regarding extraction accuracy (BER), perceptual quality (FID), and structural similarity (SSIM) are presented in Table 9.

As observed in Table 9, the proposed method exhibits robust performance across all tested resolutions. The Bit Error Rate (BER) remains consistently low ($< 0.05\%$) regardless of the image size. This confirms that our bi-directional mapping mechanism effectively synchronizes the secret message with the latent noise distribution, independent of the spatial dimensions of the latent tensor. Notably, the visual quality metrics improve as the resolution increases. The FID score decreases to 2.15 and SSIM increases to 0.96 for 256×256 images. This improvement is attributed to the inherent characteristics of the pre-trained Stable Diffusion

backbone, which is optimized for higher-resolution generation, thereby producing more coherent textures and sharper details when the latent space is larger. Conversely, the 64×64 output shows a slight degradation in FID (3.12) due to the limited pixel budget for rendering complex semantic prompts, but the steganographic integrity remains uncompromised.

In summary, these findings demonstrate that CGIS is not limited to a specific image size and can be flexibly adapted to various transmission bandwidth requirements without sacrificing security or visual fidelity.

4.10 Comparative discussion

Based on the experimental results presented above, we synthesize the findings to discuss the advantages of the CGIS framework compared to existing methods. When comparing with traditional coverless methods, CGIS shows distinct advantages. In terms of message delivery accuracy, the bi-directional mapping mechanisms in CGIS, especially the ND method, enable more precise encoding and decoding of secret information. It can effectively avoid information loss during the mapping process, which is a common problem in traditional methods. For example, in the multi-scenario transmission accuracy experiment, CGIS maintains low BER values across different image types, while traditional models like GSN and StegaDDPM have relatively higher error rates. This is because the ND method can better adapt to different

Table 8 Comparison study of three commonly used image formats

Format	BER (%)		
	1.0 bpp	3.0 bpp	6.0 bpp
PNG	3.19	3.04	2.75
JPEG	4.94	5.03	3.20
TIFF	2.98	3.01	2.68

Table 9 Performance comparison across different image resolutions

Resolution	BER (%) ↓	FID ↓	SSIM ↑
64×64	0.04	3.12	0.91
128×128 (Baseline)	0.02	2.36	0.94
256×256	0.03	2.15	0.96

data distributions, ensuring the integrity of the transmitted message.

Regarding security, the encryption-like mapping in CGIS makes it difficult for attackers to decipher the hidden information. The combination of interval-based mapping and the use of a conditional diffusion model adds an extra layer of security. In contrast, some traditional coverless methods may have vulnerabilities in information hiding, such as the susceptibility to key guessing in certain GAN-based approaches.

In terms of image quality, the GMM-ID in CGIS plays a crucial role. It can effectively reduce the impact of noise during transmission, maintaining the visual similarity between the original and stego images. As shown in the FID values in various experiments, CGIS generally has lower FID values, indicating less distortion. Traditional methods may not have effective denoising mechanisms, resulting in more obvious image quality degradation.

Furthermore, when compared with existing models such as GSN, Baseline A, and StegaDDPM, CGIS demonstrates clear advantages. It outperforms in terms of message delivery accuracy, security, and image quality across different noise levels and embedding rates. In real-world scenarios, CGIS maintains high information extraction accuracy and successfully balances message delivery, detection error, and image quality. Overall, the CGIS framework represents a significant advancement in high-quality image steganography.

Finally, regarding robustness against real-world distortions such as noise addition and JPEG compression, CGIS is designed to be resilient. The GMM-ID in CGIS plays a crucial role in this by effectively restoring the distorted image to its original or near-original state. This not only ensures accurate message extraction but also helps maintain the security of the steganographic system. For example, in the presence of Gaussian noise with different variances, CGIS still achieves relatively low BER and high image quality as measured by FID, indicating its capability to resist real-world noise attacks.

5 Conclusion

This paper proposes a novel generative steganography method capable of hiding a large amount of information in images using the diffusion model deterministic sampler CGIS. The exploration of how various factors affect the quality and accuracy of information extracted from steganography images generated by diffusion models is presented. Additionally, two binary mapping schemes tailored for the steganography of diffusion model images were designed, balancing trade-offs between image quality, detection resistance, and information accuracy. Simultaneously, the robustness of transmission is ensured by supplementing

with GMM-ID image denoising processing, demonstrating high robustness in various practical scenarios. In terms of noise immunity, this method surpasses traditional generative models and represents a breakthrough in the steganography of the diffusion model. Future work aims to enhance the steganography capability by exploring adaptive interval mapping strategies, extend the framework to support multi-modal secret information such as audio and video, and optimize computational efficiency by investigating lightweight denoising architectures.

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Data Availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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